

Predicting Late Deliveries Before They Happen

Executive Summary — APL Logistics Predictive Risk Intelligence Initiative

Prepared for: Government & Regulatory Stakeholders · APL Logistics (KWE Group)

GitHub: <https://github.com/jiteshmaurya245-dotcom/APL-Logistics-Late-Delivery-Risk>

Live App: <https://apl-logistics-late-delivery-risk-6tbthvh57qvapvpjvvtcn.streamlit.app/>

Project Links

GitHub Repository	https://github.com/jiteshmaurya245-dotcom/APL-Logistics-Late-Delivery-Risk		
Live Deployed App	https://apl-logistics-late-delivery-risk-6tbthvh57qvapvqpjvvtcn.streamlit.app/		
180,519 Orders analysed	54.8% Baseline late rate	0.762 Model accuracy (AUC)	84.3% High-risk tier precision

Purpose of This Initiative

APL Logistics (KWE Group) commissioned this initiative to address a persistent and costly challenge in global freight operations: shipments arriving late without warning. Historically, delay management has been reactive — problems are identified only after a shipment has already missed its delivery window, by which point corrective options are limited and customer impact is unavoidable.

This project introduces a predictive capability that identifies at-risk shipments before they leave the warehouse, giving operations teams the lead time needed to intervene, reroute, or proactively communicate with affected customers.

The Problem, in Plain Terms

Across 180,519 shipments analysed, more than half — 54.8% — arrived late. This is not a minor operational inefficiency; it represents systemic risk to service level agreements, financial penalties, and public trust in the logistics network. Notably, the analysis found that premium shipping options were the least reliable: orders sent via First Class experienced a 95.3% late delivery rate, compared to 38.1% for Standard Class shipments — a finding that points to a specific, addressable operational bottleneck rather than random chance.

What Was Built

A machine learning model was developed and tested against 36,104 held-out shipment records to validate its real-world accuracy before deployment. The model evaluates each shipment at the point of order — using information available before dispatch — and assigns it to one of three risk categories:

Risk Category	Share of Orders	Actual Late Rate	Recommended Response
Low Risk	27.8%	33.3%	Process normally, no action needed
Medium Risk	36.8%	43.5%	Monitor and prepare contingency
High Risk	35.1%	84.3%	Immediate intervention required

Table: Risk categories and validated outcomes

The critical measure of success is the accuracy of the High Risk category: when the system flags a shipment as high risk, it is correct 84.3% of the time — meaning operations staff can act on these alerts with a high degree of confidence, rather than chasing false alarms.

What Drives Late Deliveries

The analysis identified clear, addressable patterns behind delivery delays:

- Shipping method is the strongest single factor — First Class shipments are nearly 2.5 times more likely to be late than Standard Class, pointing to a specific fulfilment bottleneck that can be directly investigated and resolved.
- Geography is not the cause — delay rates are nearly identical (54–55%) across all five regions studied (LATAM, Europe, Pacific Asia, USCA, Africa), ruling out regional infrastructure as the primary driver.
- Order complexity matters — higher-value and higher-margin orders show elevated delay risk, suggesting these orders may require additional coordination steps that increase failure points.
- Tight schedules amplify risk — shipments scheduled for two days or less are disproportionately likely to be late, indicating a mismatch between promised delivery windows and operational capacity.

Recommended Actions

- Prioritise a review of the First Class fulfilment process — this single change addresses the largest identified risk factor.
- Deploy the predictive system into daily operations so every shipment is scored before it leaves the warehouse.
- Establish a tiered response protocol — high-risk shipments receive proactive customer outreach and expedited handling; low-risk shipments proceed without added cost or delay.
- Reassess delivery-window commitments for tight-schedule orders to ensure promises made to customers match operational reality.
- Review and retrain the model on a quarterly basis to maintain accuracy as operating conditions evolve.

Why This Matters for Public Oversight

For regulatory and government stakeholders, this initiative demonstrates a measurable, evidence-based approach to supply chain reliability — a matter of increasing public interest given the role logistics networks play in essential goods movement, economic activity, and consumer protection. Rather than relying on anecdotal reporting after service failures occur, APL Logistics can now demonstrate a quantified, auditable process for identifying and mitigating delivery risk before it affects customers.

The accompanying interactive dashboard — publicly accessible at the link on the cover page — provides full transparency into risk scoring criteria, enabling internal audit teams, regulators, or oversight bodies to review how and why any given shipment was categorised, supporting accountability and continuous improvement.

In summary: this initiative shifts APL Logistics from reactive delay management to proactive risk intelligence — using data the organisation already collects to prevent problems before they reach the customer, with a validated 84.3% accuracy rate on its highest-priority alerts.